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Artificial Intelligence and Business Analytics-
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FINAL PROJECT REPORT
Analysing the Global Interplay Among Military
Expenditures, Educational Investments,
Prosperity Rates and Happiness

Data Visualization
Prof. Dr. Johannes Reichgelt, PhD

INTRODUCTION

Governments around the world are facing a serious challenge in deciding how to divide their national budgets between military, education, and other essential public services. Military spending is important for national security, political stability, and protection from external and internal threats. Education spending, on the other hand, supports human capital, workforce skills, innovation, productivity, and long-term development. Finding the right balance between these priorities is important because both areas can be connected to a country's economic growth, prosperity, and overall well-being.

This challenge is visible across countries at different income levels. High income countries may have more financial capacity to invest in both education and military spending, while upper middle income, lower middle income, and low income countries often face more difficult budget choices. Countries such as the United States, India, Germany, Japan, and China show how national priorities can differ based on economic strength, security needs, population size, and development goals. Some countries may spend more on military because of security concerns, while others may focus more on education to support long-term social and economic progress.

This project does not try to prove that one type of spending directly causes national success. Instead, it studies how education spending and military spending are associated with GDP, GDP per capita growth, prosperity, and happiness across countries. To make the comparison more meaningful, countries are grouped using World Bank income groups: high income, upper middle income, lower middle income, and low income. This helps compare countries with more similar economic conditions instead of treating all countries as the same.

The purpose of this project is to understand how spending priorities differ across income groups and how these differences relate to development outcomes. The project uses cleaned datasets from sources such as the World Bank, SIPRI, the World Happiness Report, and the Legatum Prosperity Index. Python was used for data cleaning and preparation, while Tableau was used to create interactive dashboards, maps, trend charts, scatter plots, and parameter-based visualizations.

Through this analysis, the project aims to show patterns in education spending, military spending, economic growth, prosperity, and happiness. The goal is to provide a clear data-based view of how countries balance education and military spending, and how these choices are associated with broader national outcomes. The findings can help support better discussion around budget priorities, long-term development, and balanced public investment.

METHODOLOGY

For this project, seven cleaned datasets were used in Tableau. The datasets came from trusted sources such as the World Bank, SIPRI, the World Happiness Report, and the Legatum Prosperity Index. Python was used to clean, reshape, standardize, and prepare the raw data before importing it into Tableau. The final analysis mainly uses the period 2011 to 2021 because this is the most consistent period across education spending, military spending, GDP per capita growth, prosperity, and happiness data.

Dataset 1: Income Group Classification, Income group.csv

Source: World Bank Country and Lending Groups

<https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>

The Income group.csv file was used as the main country classification table in Tableau. It includes Country Code, Standard Country, Region, and World Bank Income Group. The income groups used in this project are high income, upper middle income, lower middle income, and low income.

This dataset was cleaned by standardizing country names and country codes so that all other datasets could be connected correctly. This table became the central table in the Tableau data model. All other datasets were connected to this file using Country Code. This helped reduce country-name mismatch problems, because country names often appear differently across different data sources.

Dataset 2: GDP Current US Dollars, GDP_Current_USD.csv

Source: World Bank GDP Current US\$

<https://data.worldbank.org/indicator/NY.GDP.MKTP.CD>

The GDP_Current_USD.csv file contains GDP values in current US dollars by country and year. This dataset was used to compare the economic size of countries and income groups. It also helped show how GDP levels differ across high income, upper middle income, lower middle income, and low income countries.

The raw World Bank file was cleaned using Python. Extra header rows and unnecessary metadata columns were removed. The data was reshaped from wide format to long format so that each row contained Country Code, Standard Country, Year, and GDP Current USD. Numeric values were cleaned and converted into proper number format. A calculated field was later created in Tableau to show GDP in billions and trillions for easier interpretation.

Dataset 3: GDP Per Capita Growth, GDP_PC_Growth.csv

Source: World Bank GDP per capita growth, annual %

<https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG>

The GDP_PC_Growth.csv file contains annual GDP per capita growth percentage by country and year. This dataset was used to measure economic growth per person rather than only total economic size. It helped compare how economic growth patterns differ across income groups from 2011 to 2021.

The raw data was cleaned using Python by removing unnecessary metadata, converting year columns into a long format, and changing values into numeric format. Country names and country codes were standardized so the dataset could connect correctly with the Income group.csv file. Rows with missing or unusable values were handled carefully to keep the data consistent for Tableau visualizations.

Dataset 4: Education Spending, Education.csv

Source: World Bank Education Expenditure Data

<https://databank.worldbank.org/source/education-statistics%3A-education-expenditure>

The Education.csv file contains education spending data by country and year. The final cleaned file includes Education Exp USD and Education Pct GDP. In this project, Education Pct GDP was mainly used because it gives a fairer comparison across countries of different economic sizes.

The raw education dataset was cleaned using Python. The original file had years spread across separate columns, so it was reshaped into long format with country, year, and education spending values. Country names and country codes were standardized. Numeric fields were cleaned, missing values were reviewed, and the final file was prepared for Tableau. Since education data coverage was strongest through 2021, education-related dashboards use the period 2011 to 2021.

Dataset 5: Military Spending, Military.csv

Source: SIPRI Military Expenditure Data through Our World in Data

<https://ourworldindata.org/grapher/military-spending-sipri>

The Military.csv file contains military expenditure data by country and year. The final cleaned file includes Military Exp USD and Military Pct GDP. In this project, Military Pct GDP was mainly used because it shows military spending as a share of each country's economy.

The raw military expenditure data was cleaned using Python. Country names and country codes were standardized, values were converted into numeric format, and unnecessary fields were removed. The dataset was reshaped and prepared so it could be connected with the Income group.csv table in Tableau. Some extreme military spending values were reviewed carefully because they can affect visual interpretation. The main comparison period used in the dashboards is 2011 to 2021.

Dataset 6: Happiness Score, Happiness.csv

Source: World Happiness Report

<https://worldhappiness.report/data-sharing/>

The Happiness.csv file contains country-level happiness data from the World Happiness Report. The main field used in this project is Happiness Score. This score was used to compare well-being outcomes across countries and income groups.

The dataset was cleaned by standardizing country names, adding or matching country codes, converting values into numeric format, and removing unnecessary fields where needed. The Happiness Score field was selected as the main happiness indicator because it had stronger coverage and was easier to compare across countries. Supporting happiness variables were not used as major analysis fields because they had more missing values. The main dashboard analysis uses 2011 to 2021 for consistency with the education and military spending data.

Dataset 7: Prosperity Score, Prosperity.csv

Source: Legatum Prosperity Index

<https://index.prosperity.com/about-prosperity/prosperity-index>

The Prosperity.csv file contains country-level Prosperity Score values from the Legatum Prosperity Index. This dataset was used to measure broader national development outcomes beyond GDP. Prosperity Score helped compare countries based on economic quality, governance, education, health, safety, and other development-related factors included in the index.

The raw Prosperity Index data was cleaned using Python. Year columns were reshaped into long format, country names were standardized, and country codes were aligned with the master country table. The final cleaned file includes Country Code, Standard Country, Year, and Prosperity Score. The analysis mainly uses the 2011 to 2021 period so that prosperity can be compared consistently with education spending, military spending, GDP per capita growth, and happiness.

After cleaning all datasets, Python was used to standardize country names, align country codes, convert values into numeric format, reshape wide datasets into long format, and remove unnecessary columns. The cleaned files were exported as CSV files and imported into Tableau.

The final Tableau workbook uses Income group.csv as the central table. The other six datasets are connected to it through Country Code. This structure allows the project to compare education spending, military spending, GDP, GDP per capita growth, happiness, and prosperity across World Bank income groups.

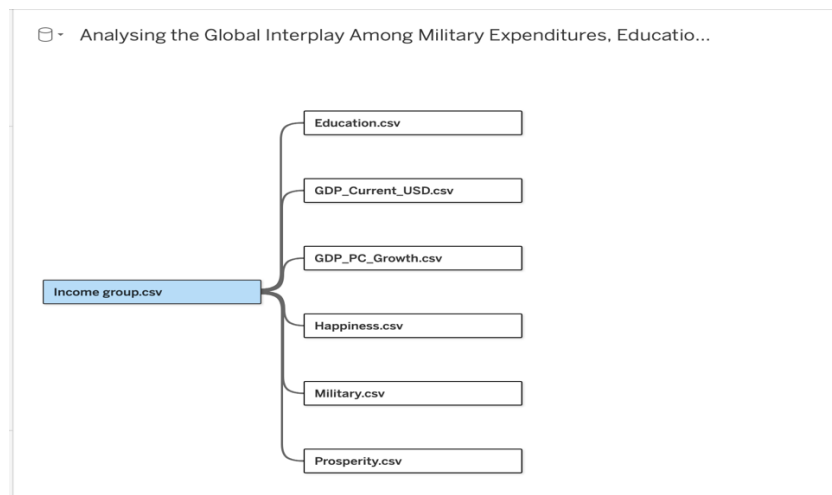
The main analysis period is 2011 to 2021 because this period gives the most consistent overlap across all key datasets. This makes the analysis cleaner, more reliable, and easier to compare across countries. The findings are interpreted as relationships and patterns, not as direct proof of cause and effect.

RESEARCH QUESTIONS

Below are the research questions that this project aims to explore:

1. How do education spending, military spending, and GDP per capita growth differ across countries by World Bank income group?
2. Is higher education spending as a percentage of GDP associated with stronger GDP per capita growth, prosperity, and happiness outcomes?
3. How is military spending as a percentage of GDP related to GDP per capita growth across different income groups?
4. Among countries with similar income levels, how do education and military spending priorities differ, and how are these priorities associated with prosperity and happiness outcomes?
5. Which countries stand out as high education spenders, high military spenders, high growth countries, or prosperity and happiness outliers?

DATA MODEL AND TABLE RELATIONSHIPS



In this project, seven cleaned datasets are used to build the Tableau workbook. The data model is centered on `Income group.csv`, which works as the main country classification table. This table is connected to all other datasets through Country Code, which helps reduce country-name mismatch issues.

The connected tables include `Education.csv`, `GDP_Current_USD.csv`, `GDP_PC_Growth.csv`, `Happiness.csv`, `Military.csv`, and `Prosperity.csv`. These links allow the workbook to combine education spending, military spending, GDP, GDP per capita growth, happiness scores, prosperity scores, region, and World Bank income group in one analysis.

This structure supports different types of visualizations, including trend charts, scatter plots, bar charts, maps, Top 10 country rankings, and parameter-based dashboards. It also allows countries to be compared by World Bank income group: high income, upper middle income, lower middle income, and low income.

This data model provides a clean and consistent structure for analyzing how education and military spending are associated with GDP growth, prosperity, and happiness outcomes across countries. Since the datasets are connected using Country Code, the workbook is more reliable and easier to use in Tableau.

ANALYSIS OF THE VISUALIZATIONS:-

1. GDP Trend by World Bank Income Group, 2011 to 2021

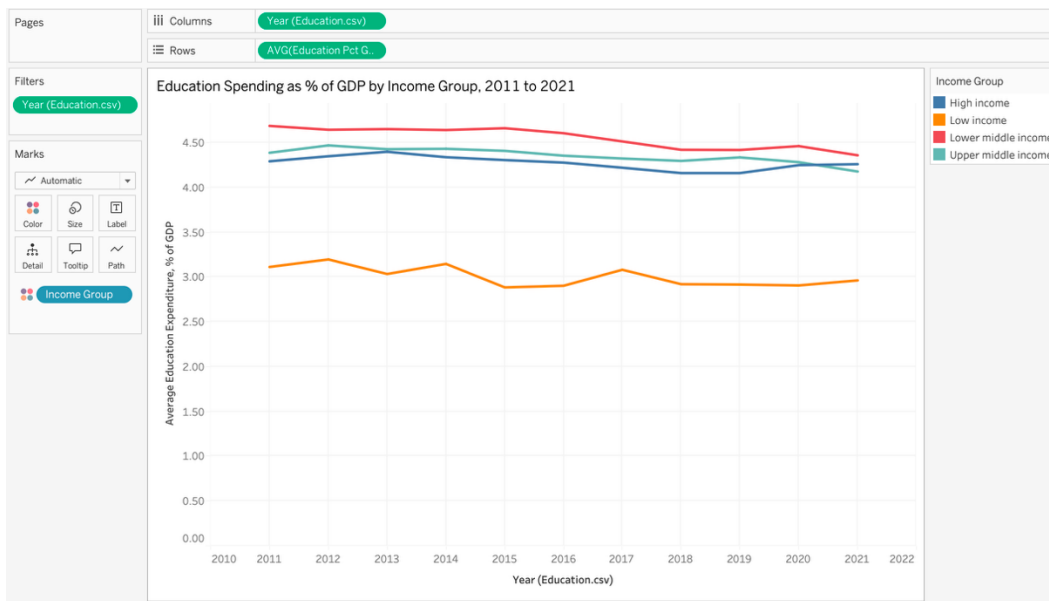


This chart shows the GDP trend by World Bank income group from 2011 to 2021. The line chart compares high income, upper middle income, lower middle income, and low income countries using average GDP in trillions.

High income countries have the highest average GDP throughout the period. Their GDP remains much higher than all other income groups and shows a clear increase after 2016, reaching the highest level in 2021. Upper middle income countries also show steady growth, especially after 2016, but they remain below high income countries. Lower middle income countries show slower but consistent growth over time. Low income countries remain at the lowest level throughout the period, with only a small increase.

The chart clearly shows a large economic gap between income groups. It supports **Research Question 1** by showing how GDP trends differ across World Bank income groups. The graph also shows why income group classification is important for this project, because countries at different income levels have very different economic capacity to invest in education, military, and other development priorities.

2. Education Spending as % of GDP by Income Group, 2011 to 2021

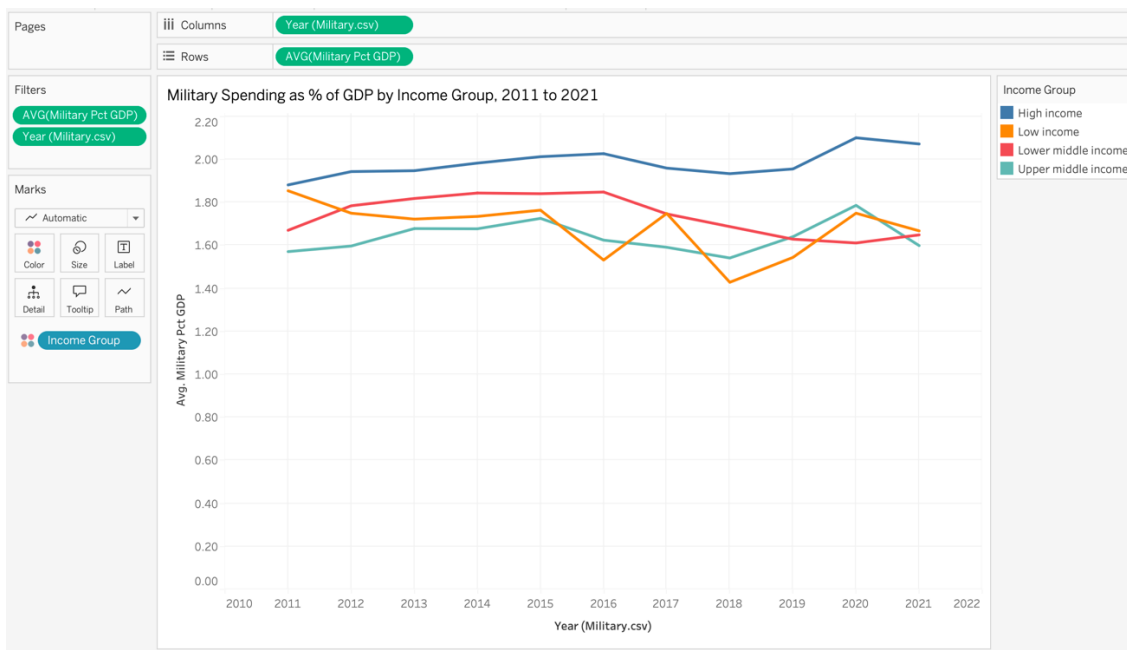


This chart shows education spending as a percentage of GDP by World Bank income group from 2011 to 2021. The line chart compares high income, upper middle income, lower middle income, and low income countries using average education expenditure as a percentage of GDP.

Lower middle income countries show the highest average education spending for most of the period. Their education spending stays around 4.4% to 4.7% of GDP, although it slightly declines after 2016. Upper middle income countries and high income countries remain close to each other, mostly between 4.2% and 4.5% of GDP. Low income countries have the lowest education spending, staying around 2.9% to 3.2% of GDP throughout the period.

The chart shows that education spending differs clearly across income groups. It also shows that higher income does not always mean a higher percentage of GDP spent on education. Some lower middle income countries spend a larger share of GDP on education than high income countries. This graph supports **Research Question 1** by showing how education spending patterns vary across World Bank income groups over time.

3. Military Spending as % of GDP by Income Group, 2011 to 2021

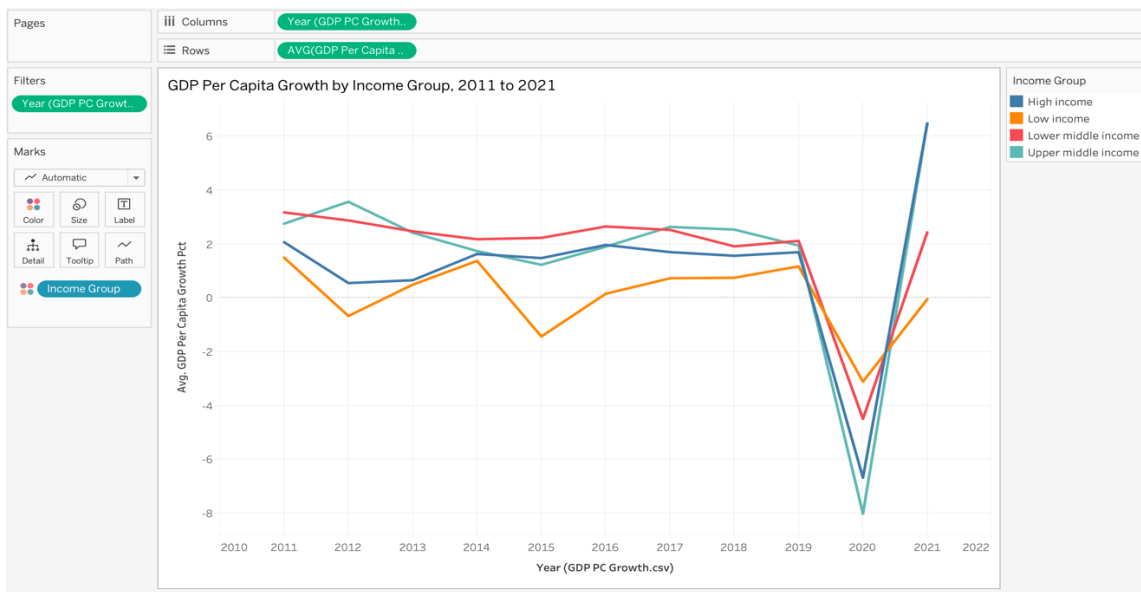


This chart shows military spending as a percentage of GDP by World Bank income group from 2011 to 2021. The line chart compares high income, upper middle income, lower middle income, and low income countries using average military expenditure as a percentage of GDP.

High income countries have the highest military spending for most of the period. Their spending stays close to 1.9% to 2.1% of GDP and increases slightly after 2019. Lower middle income countries remain mostly stable, staying around 1.6% to 1.8% of GDP. Upper middle income countries show some fluctuation, with spending rising around 2020 and then slightly declining in 2021. Low income countries show the most variation, with spending dropping in 2016 and 2018, then increasing again in 2020 before falling slightly in 2021.

The graph shows that military spending patterns differ across income groups. High income countries spend the highest average share of GDP on military, while low income countries show more fluctuation over time. This graph supports **Research Question 1** by showing how military spending differs across World Bank income groups. It also supports **Research Question 3** by helping compare military spending patterns with economic growth across income groups.

4. GDP Per Capita Growth by Income Group, 2011 to 2021



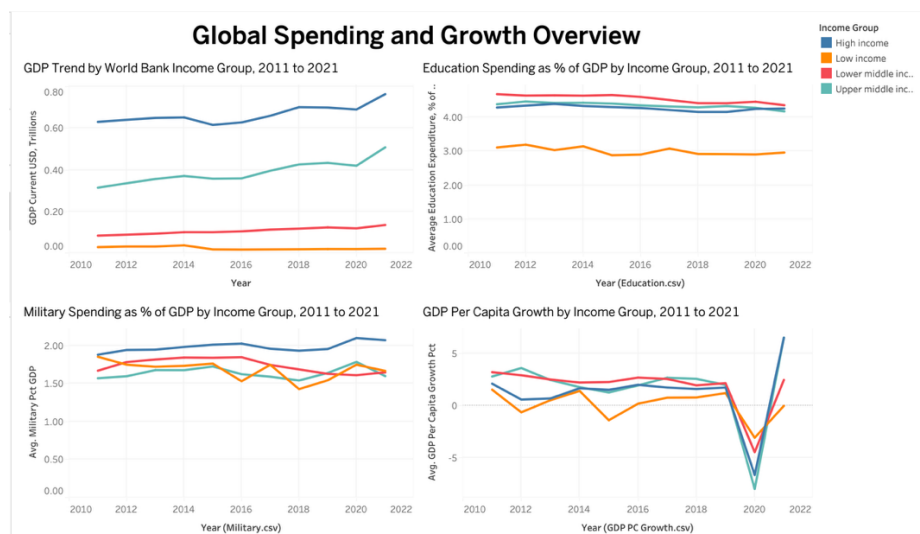
This chart shows GDP per capita growth by World Bank income group from 2011 to 2021. The line chart compares high income, upper middle income, lower middle income, and low income countries using average GDP per capita growth percentage.

From 2011 to 2019, most income groups show positive GDP per capita growth, but the level of growth is different. Lower middle income and upper middle income countries generally show stronger growth than high income countries during many years. Low income countries show more fluctuation, with some years of positive growth and some years of negative growth.

The graph also shows a sharp decline in 2020 for all income groups. This likely reflects the global economic slowdown during the COVID-19 period. Upper middle income and high income countries show the deepest decline in 2020, while low income countries also fall below zero. In 2021, all groups recover, with high income countries showing the strongest rebound.

The graph shows that GDP per capita growth changes differently across income groups. It supports **Research Question 1** by showing how economic growth patterns vary by World Bank income group. The chart also shows that growth is not always steady, and global shocks can affect all income groups at the same time.

Dashboard 1 : Global Spending and Growth Overview



This dashboard explores how GDP, education spending, military spending, and GDP per capita growth differ across World Bank income groups from 2011 to 2021. The four visualizations collectively address Research Question 1:

How do education spending, military spending, and GDP per capita growth differ across countries by World Bank income group?

It enables users to:

- Compare GDP trends across high income, upper middle income, lower middle income, and low income countries.
- Understand how education spending as a percentage of GDP differs across income groups.
- Compare military spending as a percentage of GDP across income groups.
- Analyze how GDP per capita growth changes across income groups over time.

The dashboard shows that high income countries have the highest average GDP throughout the period, followed by upper middle income countries. Lower middle income and low income countries remain much lower in terms of average GDP, showing a clear economic gap between income groups.

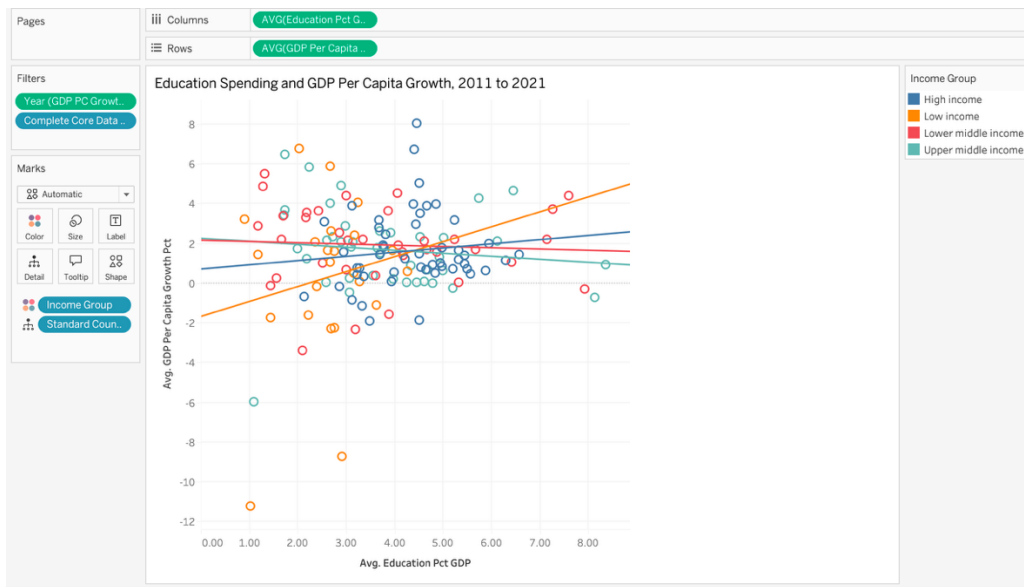
In education spending, lower middle income countries spend the highest average share of GDP on education for most years, while low income countries spend the lowest. This shows that higher income does not always mean a higher percentage of GDP is spent on education.

In military spending, high income countries show the highest average military spending as a percentage of GDP. Other income groups show more fluctuation, especially low income and upper middle income countries.

The GDP per capita growth chart shows that growth is not stable across all years. Most groups show positive growth from 2011 to 2019, followed by a sharp decline in 2020 and recovery in 2021. This shows how global shocks affected all income groups, although the level of decline and recovery differed.

The dashboard clearly answers **Research Question 1** by showing that GDP, education spending, military spending, and GDP per capita growth vary across World Bank income groups. It also shows why income group classification is important, because countries with different income levels have different economic capacity and spending patterns.

5. Education Spending and GDP Per Capita Growth, 2011 to 2021



This chart shows the relationship between education spending as a percentage of GDP and GDP per capita growth from 2011 to 2021. Each point represents a country, and the colors show World Bank income groups. The trend lines help compare how this relationship differs across income groups.

The scatter plot shows a mixed relationship between education spending and GDP per capita growth. Low income countries show a stronger upward trend, meaning that higher education spending is associated with higher GDP per capita growth in this group. High income countries also show a slight positive trend, but the relationship is weaker. Lower middle income countries have a nearly flat trend, while upper middle income countries show a slight negative trend.

This chart supports **Research Question 2** by showing that education spending is related to GDP per capita growth, but the relationship is not the same for every income group. The chart suggests that education spending alone does not fully explain economic growth. Other factors such as governance, infrastructure, policy quality, and economic conditions may also influence how education spending connects with growth.

6. Education Spending and Prosperity Score, 2011 to 2021

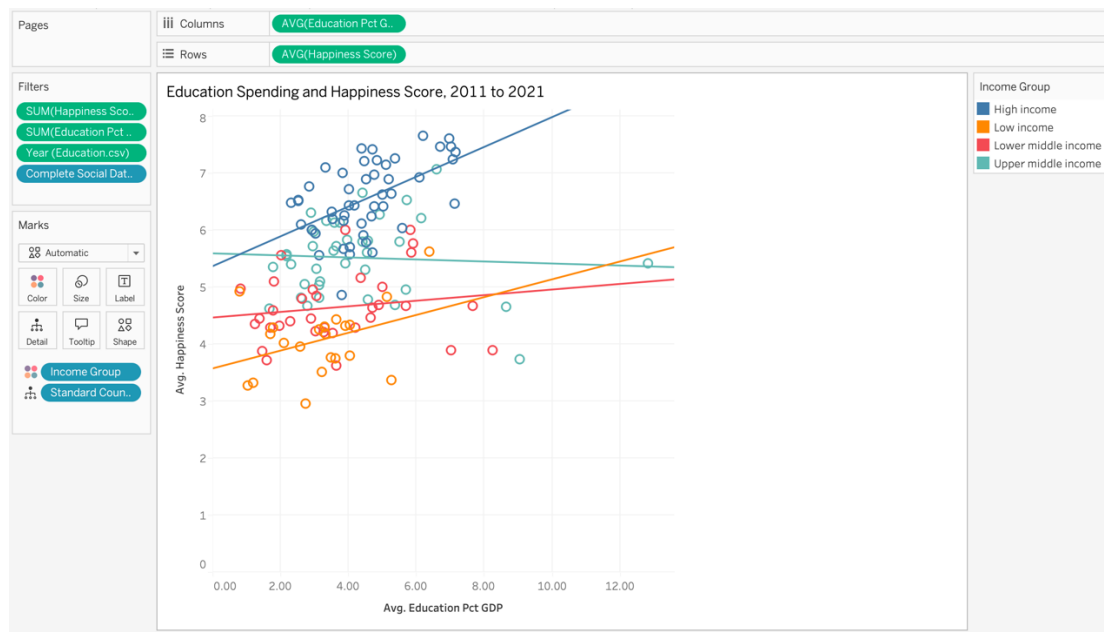


This chart shows the relationship between education spending as a percentage of GDP and Prosperity Score from 2011 to 2021. Each point represents a country, and the colors show World Bank income groups. The trend lines show how the relationship between education spending and prosperity differs across income groups.

The scatter plot shows that high income countries generally have the highest Prosperity Scores, mostly above 60, even when their education spending levels vary. Low income countries have the lowest Prosperity Scores, mostly between 30 and 50, but their trend line moves upward, showing that higher education spending is associated with higher prosperity within this group. Upper middle income and lower middle income countries show a weaker relationship, with more variation across countries.

This chart supports **Research Question 2** by showing that education spending is associated with prosperity outcomes, but the strength of the relationship differs by income group. The chart also shows that education spending alone does not fully explain prosperity. Income level, governance, health, infrastructure, and policy quality may also affect how education spending connects with national prosperity.

7. Education Spending and Happiness Score, 2011 to 2021

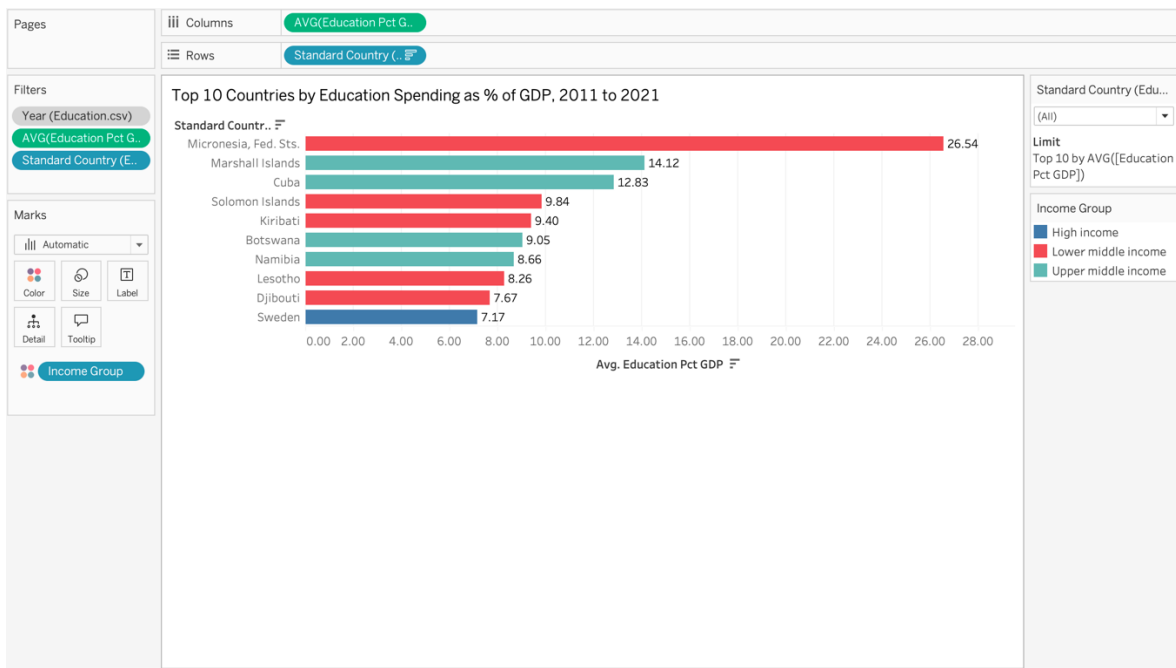


This chart shows the relationship between education spending as a percentage of GDP and Happiness Score from 2011 to 2021. Each point represents a country, and the colors show the World Bank income groups. The trend lines help compare how the relationship between education spending and happiness differs across income groups.

The scatter plot shows that high income countries generally have the highest Happiness Scores, mostly between 6 and 8. Low income countries usually have lower Happiness Scores, mostly between 3 and 5. The trend line for high income countries moves upward, showing a positive relationship between education spending and happiness. Low income and lower middle income countries also show a slight positive trend, while upper middle income countries show a flatter trend.

This chart supports **Research Question 2** by showing that education spending is associated with happiness outcomes, but the relationship is not the same for every income group. Higher education spending may be connected with better well-being, but happiness is also affected by income level, health, social support, governance, safety, and quality of public services.

8. Top 10 Countries by Education Spending as % of GDP, 2011 to 2021

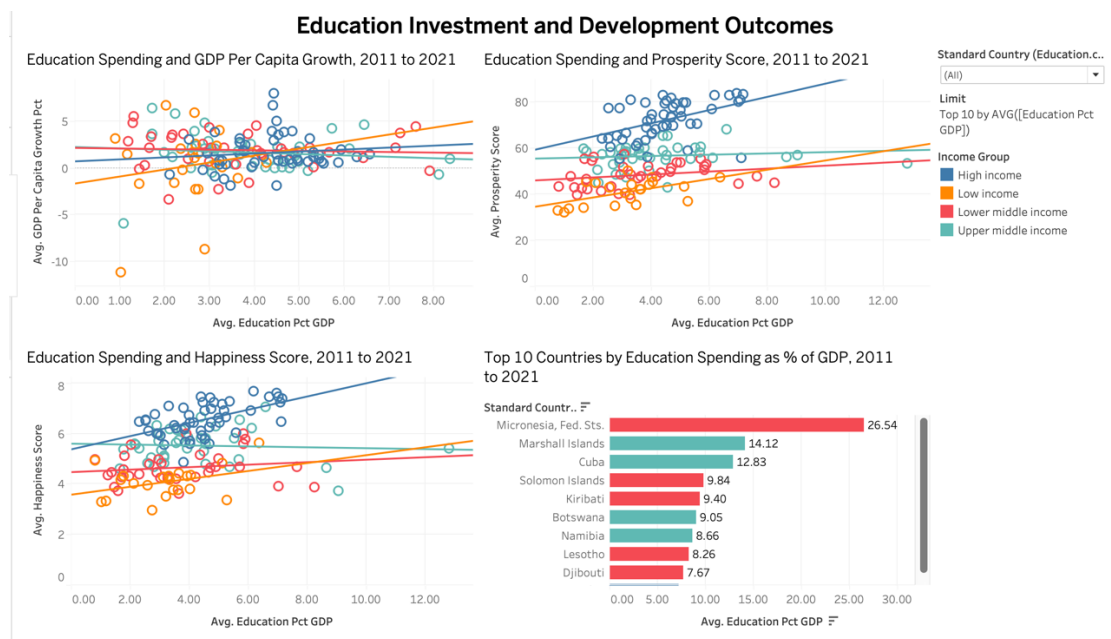


This chart shows the Top 10 countries by average education spending as a percentage of GDP from 2011 to 2021. The bar chart compares countries across World Bank income groups and highlights which countries spend the highest share of their GDP on education.

Micronesia, Fed. Sts. has the highest average education spending at 26.54% of GDP, which is much higher than all other countries in the chart. Marshall Islands and Cuba also show very high education spending, followed by Solomon Islands, Kiribati, Botswana, Namibia, Lesotho, Djibouti, and Sweden. Most of the top countries are lower middle income or upper middle income countries, while Sweden is the only high income country in this Top 10 list.

The chart shows that high education spending as a percentage of GDP is not limited to high income countries. Some lower and upper middle income countries allocate a larger share of GDP to education than many richer countries. This chart supports **Research Question 2** by showing which countries prioritize education spending the most during 2011 to 2021.

Dashboard 2 : Education Investment and Development Outcomes



This dashboard explores how education spending is associated with development outcomes across World Bank income groups from 2011 to 2021. The four visualizations collectively address Research Question 2:

Is higher education spending as a percentage of GDP associated with stronger GDP per capita growth, prosperity, and happiness outcomes?

It enables users to:

- Assess the relationship between education spending and GDP per capita growth.
- Explore how education spending is associated with Prosperity Score across income groups.
- Compare the relationship between education spending and Happiness Score.
- Identify the top 10 countries that spend the highest average share of GDP on education.

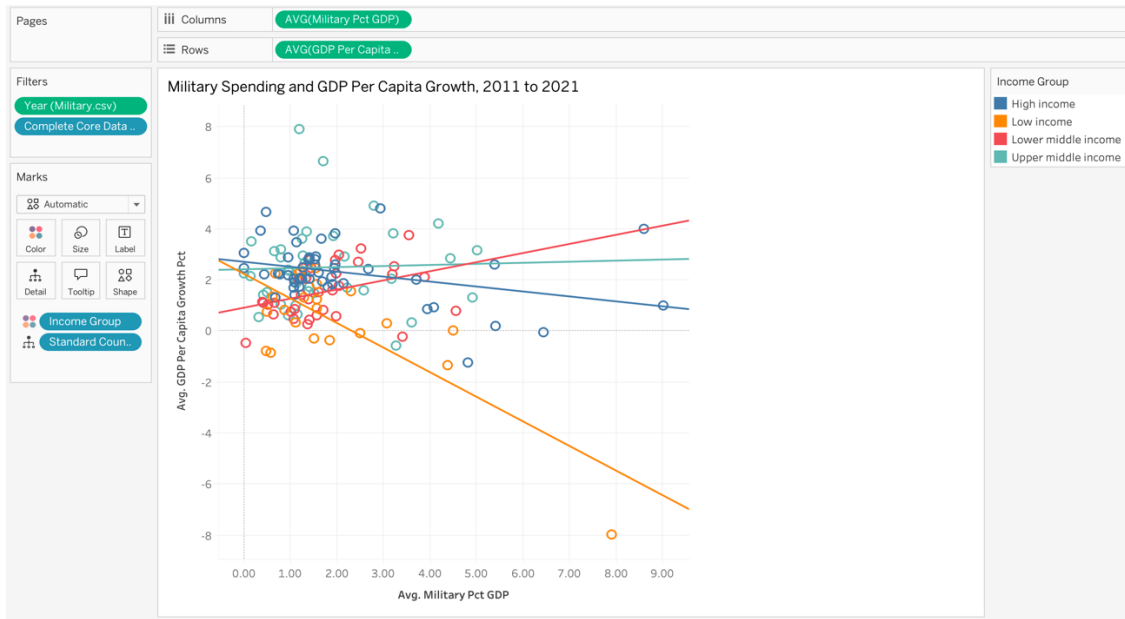
The dashboard shows that education spending has a mixed relationship with GDP per capita growth. Some income groups show a positive trend, while others show weaker or uneven patterns. This means higher education spending does not automatically translate into stronger short-term economic growth.

The prosperity and happiness charts show a clearer pattern. High income countries generally have higher prosperity and happiness scores, while low income countries remain lower. In some groups, higher education spending is associated with better prosperity and happiness outcomes, but the relationship differs by income level.

The Top 10 education spending chart shows that some lower middle income and upper middle income countries spend a very high percentage of GDP on education. This shows that education priority is not limited to rich countries.

The dashboard supports Research Question 2 by showing that education spending is associated with development outcomes, especially prosperity and happiness. However, the relationship with GDP per capita growth is mixed. The dashboard suggests that education spending matters, but outcomes also depend on income level, governance, infrastructure, policy quality, and other country-specific factors.

9. Military Spending and GDP Per Capita Growth, 2011 to 2021

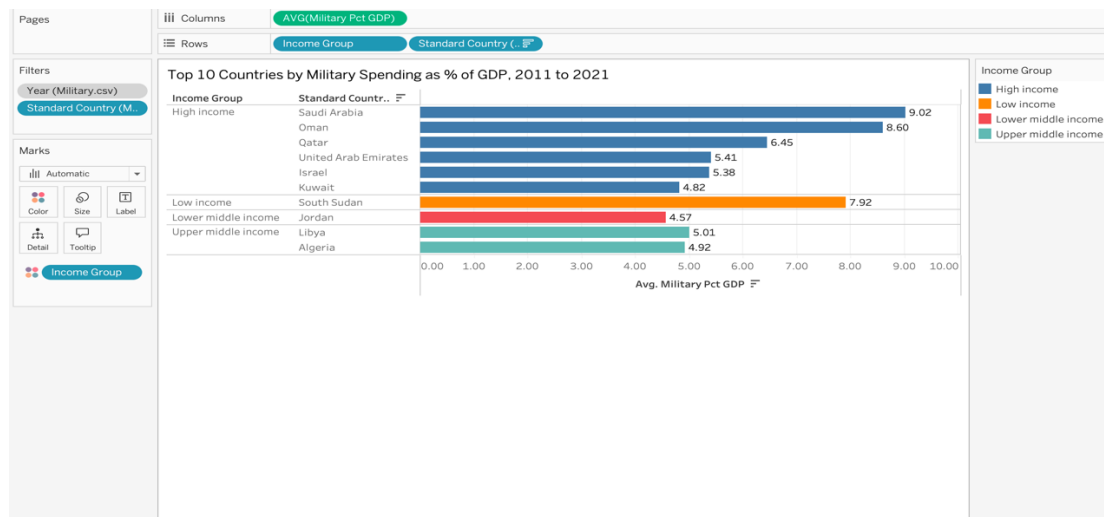


This chart shows the relationship between military spending as a percentage of GDP and GDP per capita growth from 2011 to 2021. Each point represents a country, and the colors show World Bank income groups. The trend lines show how the relationship between military spending and growth differs across income groups.

The scatter plot shows mixed patterns across income groups. High income countries show a slightly negative trend, meaning higher military spending is not associated with higher GDP per capita growth in this group. Low income countries show a stronger negative trend, with some countries having higher military spending and lower growth. Lower middle income and upper middle income countries show a positive trend, but the points are spread out, so the relationship is not very strong.

The chart supports **Research Question 3** by showing that military spending is related to GDP per capita growth differently across income groups. The relationship is not consistent, so it would not be correct to say that military spending directly increases or reduces growth. The chart suggests that other factors, such as conflict, governance, economic stability, and investment quality, may also affect how military spending connects with economic growth.

10. Top 10 Countries by Military Spending as % of GDP, 2011 to 2021

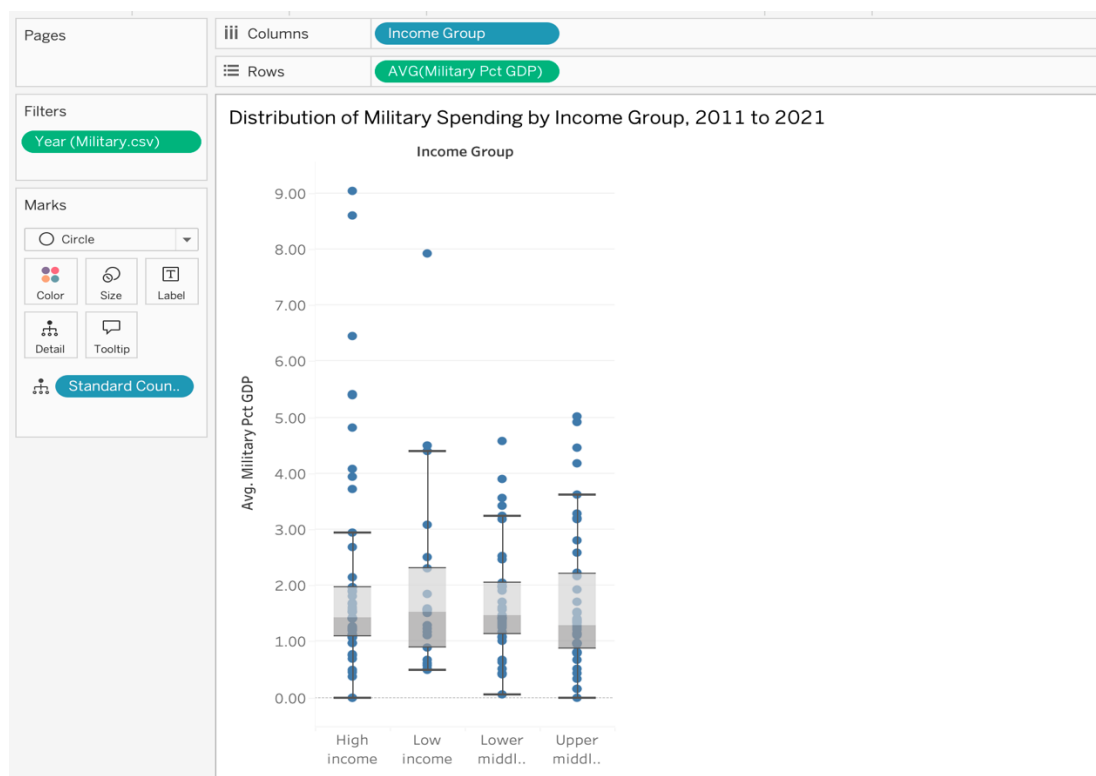


This chart shows the Top 10 countries by average military spending as a percentage of GDP from 2011 to 2021. The bar chart compares countries across World Bank income groups and highlights which countries spend the highest share of their GDP on military spending.

Saudi Arabia has the highest average military spending at 9.02% of GDP, followed by Oman at 8.60% and South Sudan at 7.92%. Qatar, United Arab Emirates, Israel, Libya, Algeria, Kuwait, and Jordan also appear among the top military spenders. Most of the top countries are high income or upper middle income countries, with South Sudan appearing as the main low income country in the list.

The above chart supports **Research Question 3** by showing which countries allocate the largest share of GDP to military spending. The chart also shows that high military spending is not limited to one income group, although several high income countries appear at the top. This helps compare military priorities across countries and income groups without claiming that military spending directly causes economic growth.

11. Distribution of Military Spending by Income Group, 2011 to 2021

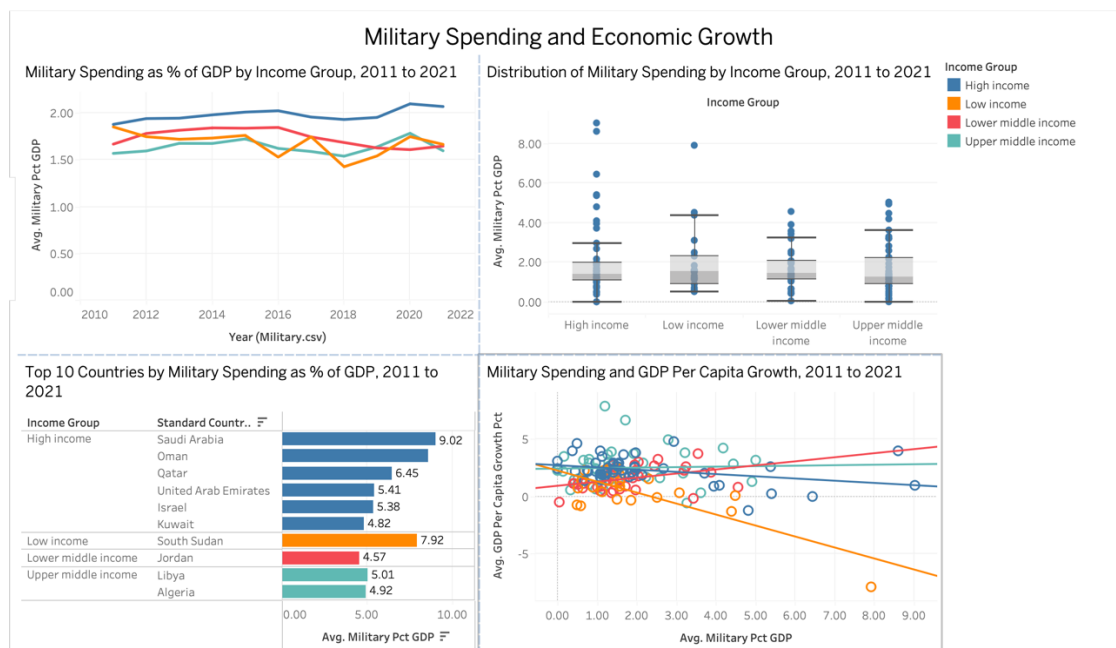


This chart shows the distribution of military spending as a percentage of GDP by World Bank income group from 2011 to 2021. The chart compares high income, low income, lower middle income, and upper middle income countries. Each point represents a country, and the box plot shows the spread of military spending within each income group.

High income countries show several high military spending outliers, with some countries spending above 8% or 9% of GDP. Low income and upper middle income countries also show wide variation, meaning military spending differs strongly within these groups. Lower middle income countries have a more moderate spread, though some countries still spend above the average level.

The chart supports **Research Question 3** by showing that military spending is not the same across income groups or even within the same income group. Some countries spend a much higher share of GDP on military than others, which shows why country-level outliers are important. The chart also supports the idea that military spending patterns should be interpreted carefully, because income group alone does not fully explain military spending behavior.

Dashboard 3 : Military Spending and Economic Growth



This dashboard explores how military spending as a percentage of GDP is related to GDP per capita growth across World Bank income groups from 2011 to 2021. The four visualizations collectively address Research Question 3:

How is military spending as a percentage of GDP related to GDP per capita growth across different income groups?

It enables users to:

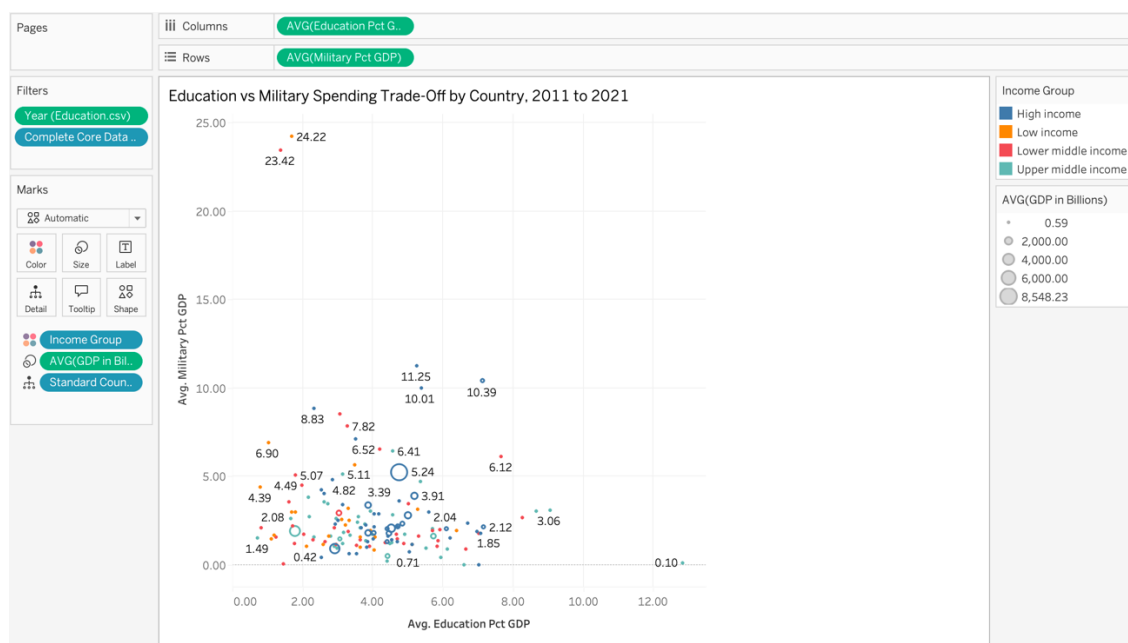
- Compare military spending trends across high income, upper middle income, lower middle income, and low income countries.
- Understand how military spending is distributed within each income group.
- Identify the top 10 countries with the highest military spending as a percentage of GDP.
- Analyze the relationship between military spending and GDP per capita growth across income groups.

The dashboard shows that high income countries generally have the highest average military spending as a percentage of GDP during the period. The distribution chart also shows that military spending varies widely within income groups, with several high-spending outliers. The Top 10 chart shows that countries such as Saudi Arabia, Oman, South Sudan, Qatar, and the United Arab Emirates spend the highest share of GDP on military.

The scatter plot shows that the relationship between military spending and GDP per capita growth is mixed. Some income groups show a positive trend, while others show a negative trend. This means higher military spending is not consistently associated with stronger economic growth.

The dashboard supports **Research Question 3** by showing that military spending patterns differ across income groups and countries. It also shows that military spending alone does not explain GDP per capita growth. Other factors such as conflict, governance, economic stability, natural resources, and policy decisions may also affect economic outcomes.

12. Education vs Military Spending Trade-Off by Country, 2011 to 2021

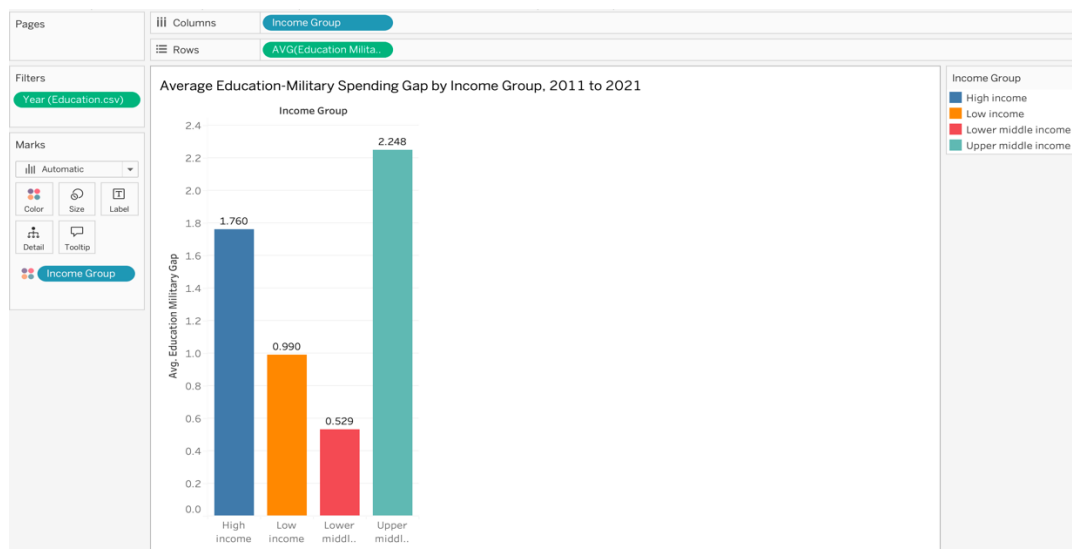


This chart shows the education versus military spending trade-off by country from 2011 to 2021. The x-axis shows average education spending as a percentage of GDP, and the y-axis shows average military spending as a percentage of GDP. Each point represents a country, the color shows its World Bank income group, and the bubble size shows average GDP in billions.

Most countries are clustered in the lower-left area of the chart, meaning they spend a moderate share of GDP on both education and military. A few countries stand out with very high military spending, especially the points near the top of the chart. Some countries also show higher education spending with lower military spending, especially toward the right side of the chart. This shows that countries do not follow the same spending pattern, even when they belong to similar income groups.

The above chart supports **Research Question 4** by showing how education and military spending priorities differ across countries. It helps identify whether countries are more education-focused, military-focused, or balanced. The chart also shows that income group alone does not fully explain spending choices, because countries within the same income group can still have very different budget priorities.

13. Average Education-Military Spending Gap by Income Group, 2011 to 2021

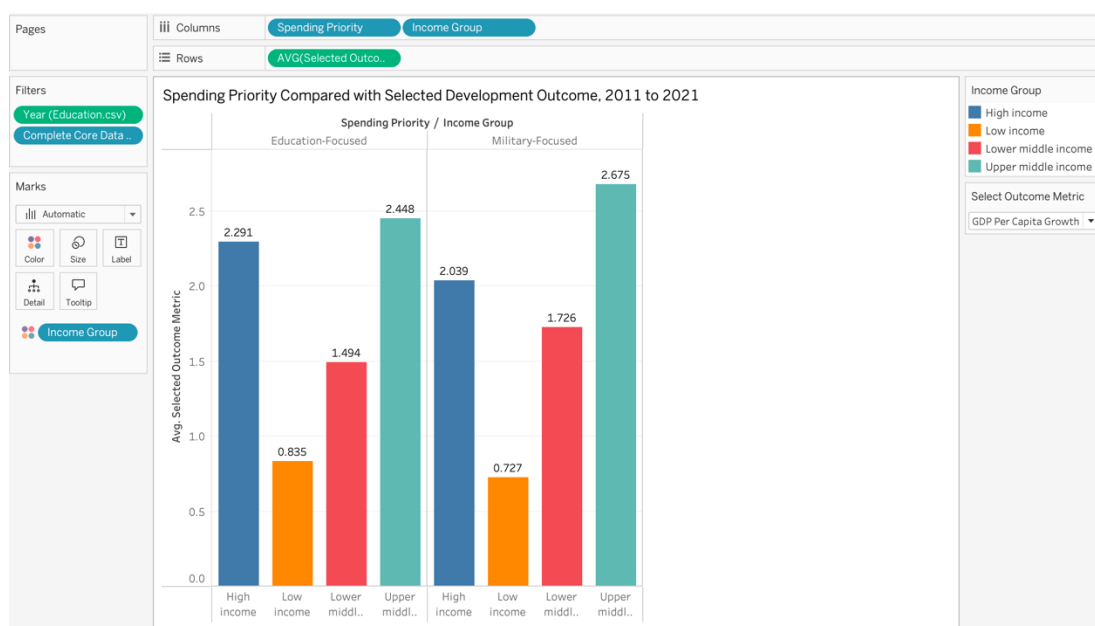


This chart shows the average education-military spending gap by World Bank income group from 2011 to 2021. The gap is calculated by subtracting military spending as a percentage of GDP from education spending as a percentage of GDP. A higher positive value means that the income group spends a larger share of GDP on education compared with military expenditure.

Upper middle income countries show the highest average gap at 2.248 percentage points, followed by high income countries at 1.760 percentage points. Low income countries also spend more on education than military, but the gap is smaller at 0.990 percentage points. Lower middle income countries have the smallest gap at 0.529 percentage points, meaning education and military spending are closer in this group.

The bar graph supports **Research Question 4** by showing how spending priorities differ across income groups. All groups show a positive gap, meaning education spending is higher than military spending on average. However, the size of the gap differs, showing that countries at different income levels balance education and military spending differently.

14. Spending Priority Compared with Selected Development Outcome, 2011 to 2021

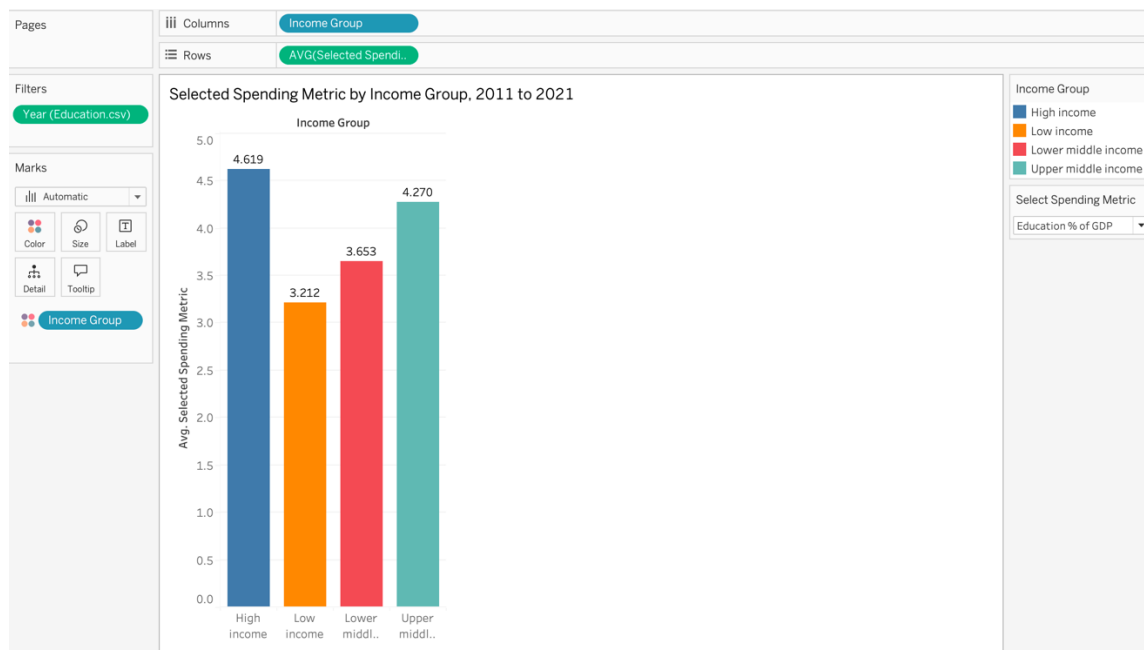


This chart compares spending priority with the selected development outcome from 2011 to 2021. In the current view, the selected outcome metric is GDP per capita growth. The chart separates countries into education-focused and military-focused groups, then compares the average outcome across World Bank income groups.

The chart shows that upper middle income countries have the highest average GDP per capita growth in both education-focused and military-focused groups. High income countries also show positive growth, but the average is lower than upper middle income countries. Low income countries show the lowest average growth in both spending priority groups. Lower middle income countries show moderate growth, with slightly higher growth in military-focused countries than education-focused countries.

The bar graph supports **Research Question 4** by showing how different spending priorities are associated with development outcomes across income groups. It shows that spending priority alone does not fully explain GDP per capita growth, because income group differences are still clearly visible. The parameter makes the chart more useful because the outcome can be changed to GDP per capita growth, Prosperity Score, or Happiness Score, allowing the same chart to compare different development outcomes without creating extra charts.

15. Selected Spending Metric by Income Group, 2011 to 2021

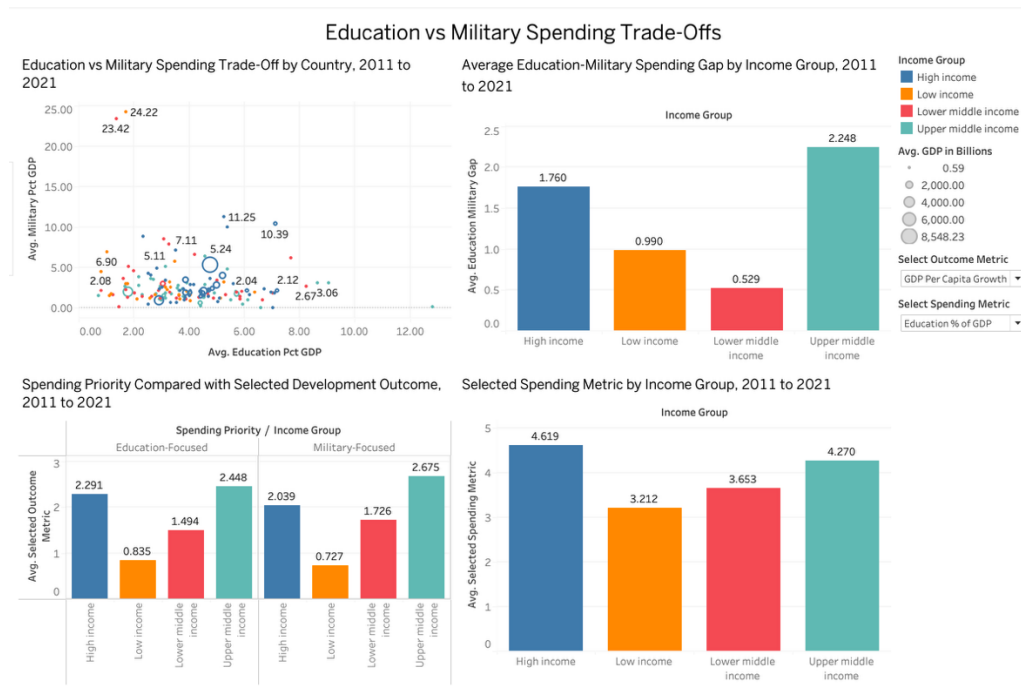


This chart shows the selected spending metric by World Bank income group from 2011 to 2021. In the current view, the selected spending metric is Education % of GDP. The chart compares the average education spending share across high income, upper middle income, lower middle income, and low income countries.

High income countries show the highest average education spending at 4.619% of GDP, followed by upper middle income countries at 4.270%. Lower middle income countries spend around 3.653%, while low income countries show the lowest average at 3.212%. This shows that education spending as a share of GDP differs clearly across income groups.

The bar chart supports **Research Question 4** by allowing users to compare spending priorities across income groups. Since the chart uses a parameter, the selected metric can be changed to Education % of GDP, Military % of GDP, or Education-Military Gap. This makes the chart useful for comparing how different income groups balance education and military spending without creating separate charts for each metric.

Dashboard 4 : Education vs Military Spending Trade-Offs



This dashboard explores how education and military spending priorities differ across World Bank income groups from 2011 to 2021. The four visualizations collectively address Research Question 4:

Among countries with similar income levels, how do education and military spending priorities differ, and how are these priorities associated with prosperity and happiness outcomes?

It enables users to:

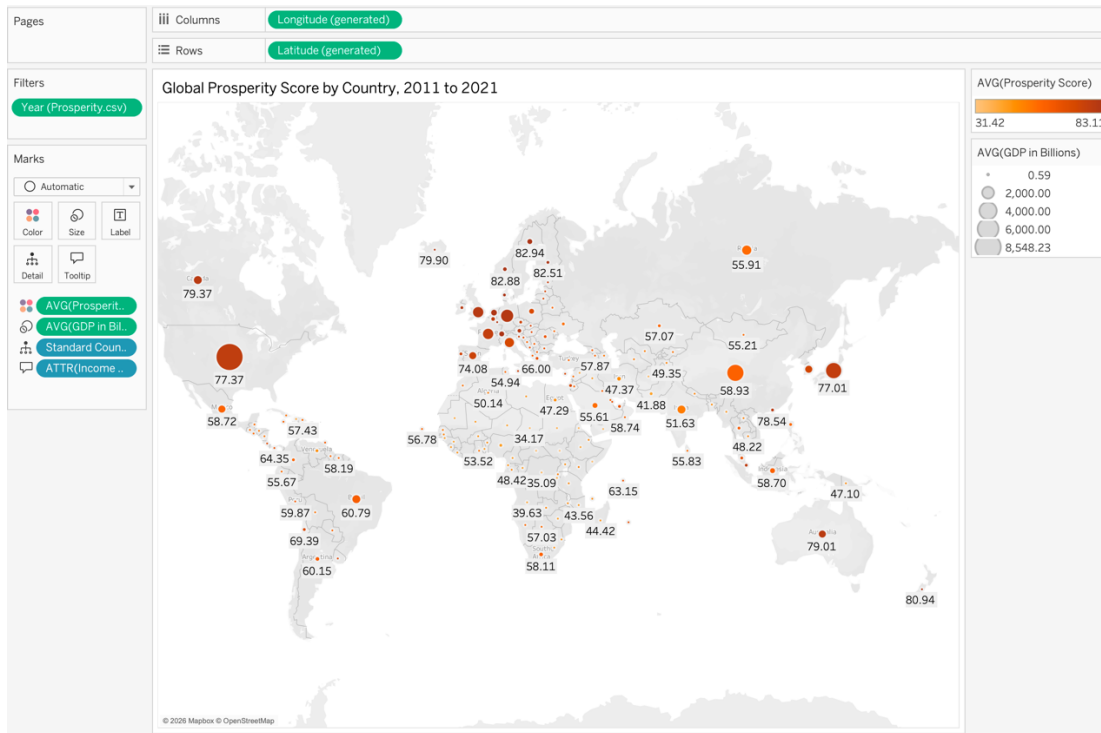
- Compare education spending and military spending as a percentage of GDP at the country level.
- Understand the average gap between education and military spending across income groups.
- Compare spending priorities, such as education-focused and military-focused countries, with selected development outcomes.
- Use the spending metric parameter to switch between education spending, military spending, and the education-military spending gap.

The dashboard shows that most countries spend a moderate share of GDP on both education and military expenditure, but some countries stand out with very high military spending or very high education spending. The education-military gap chart shows that all income groups spend more on education than military on average, but the size of the gap differs. Upper middle income countries show the largest education-military gap, while lower middle income countries show the smallest gap.

The parameter-based charts make the dashboard more interactive. The selected outcome chart allows the user to compare spending priorities with GDP per capita growth, prosperity, or happiness. The selected spending metric chart allows the user to compare education spending, military spending, or the spending gap by income group.

This dashboard supports **Research Question 4** by showing that countries within the same income group can still follow different spending priorities. It also shows that spending priorities are associated with different development outcomes, but the dashboard does not prove direct cause and effect. Other factors such as governance, conflict, economic stability, health systems, and policy quality may also influence prosperity and happiness outcomes.

16. Global Prosperity Score by Country, 2011 to 2021

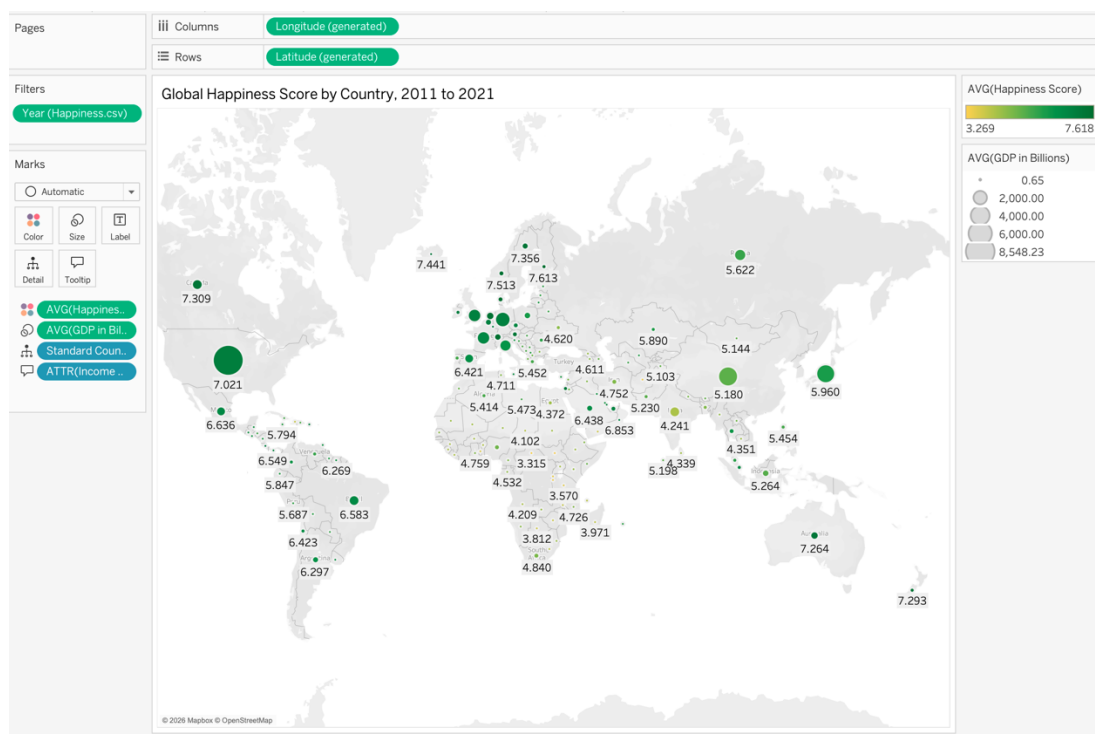


This map shows the average Prosperity Score by country from 2011 to 2021. Each circle represents a country. The color shows the average Prosperity Score, and the size of the circle shows average GDP in billions. Larger circles represent countries with larger economies, while darker colors represent higher Prosperity Scores.

The map shows that countries in North America, Western Europe, Australia, New Zealand, and some parts of East Asia have higher Prosperity Scores. Many countries in parts of Africa and South Asia show lower Prosperity Scores. The United States, China, and Japan appear as larger circles because of their larger GDP size, but the map also shows that economic size alone does not fully explain prosperity.

This chart supports **Research Question 5** by helping identify prosperity outliers across countries. It shows that prosperity differs strongly by region and country, and that some countries with large economies do not always have the highest prosperity scores. This helps compare country-level outcomes beyond only GDP and spending levels.

17. Global Happiness Score by Country, 2011 to 2021

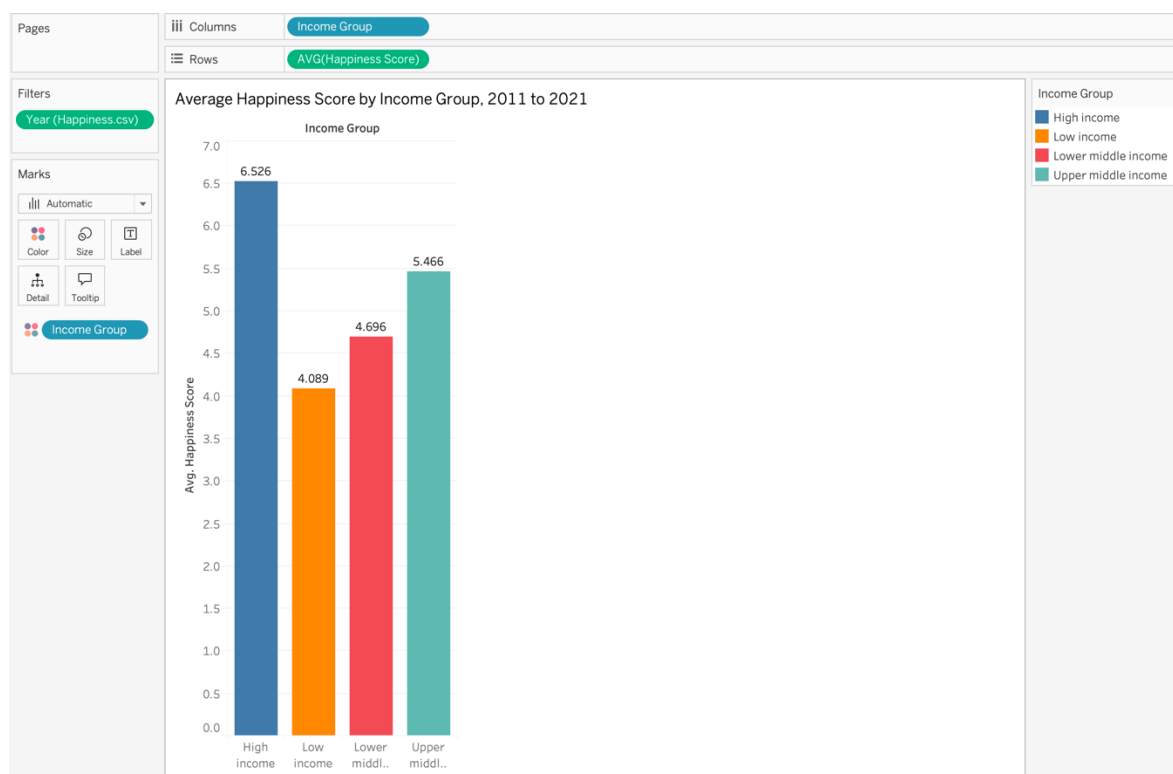


This map shows the average Happiness Score by country from 2011 to 2021. Each circle represents a country. The color shows the average Happiness Score, and the size of the circle shows average GDP in billions. Larger circles represent countries with larger economies, while darker green circles show higher happiness scores.

The map shows that countries in North America, Western Europe, Australia, New Zealand, and some parts of Northern Europe have higher Happiness Scores. Many countries in parts of Africa, South Asia, and the Middle East show lower scores. The United States, China, and Japan appear as larger circles because of their larger GDP size, but their happiness levels are not the same, which shows that GDP size alone does not explain happiness.

This chart supports **Research Question 5** by helping identify happiness outliers across countries. It shows that happiness differs strongly by country and region, and that some countries with large economies do not always have the highest happiness scores. This helps compare country-level well-being beyond GDP, education spending, and military spending.

18. Average Happiness Score by Income Group, 2011 to 2021

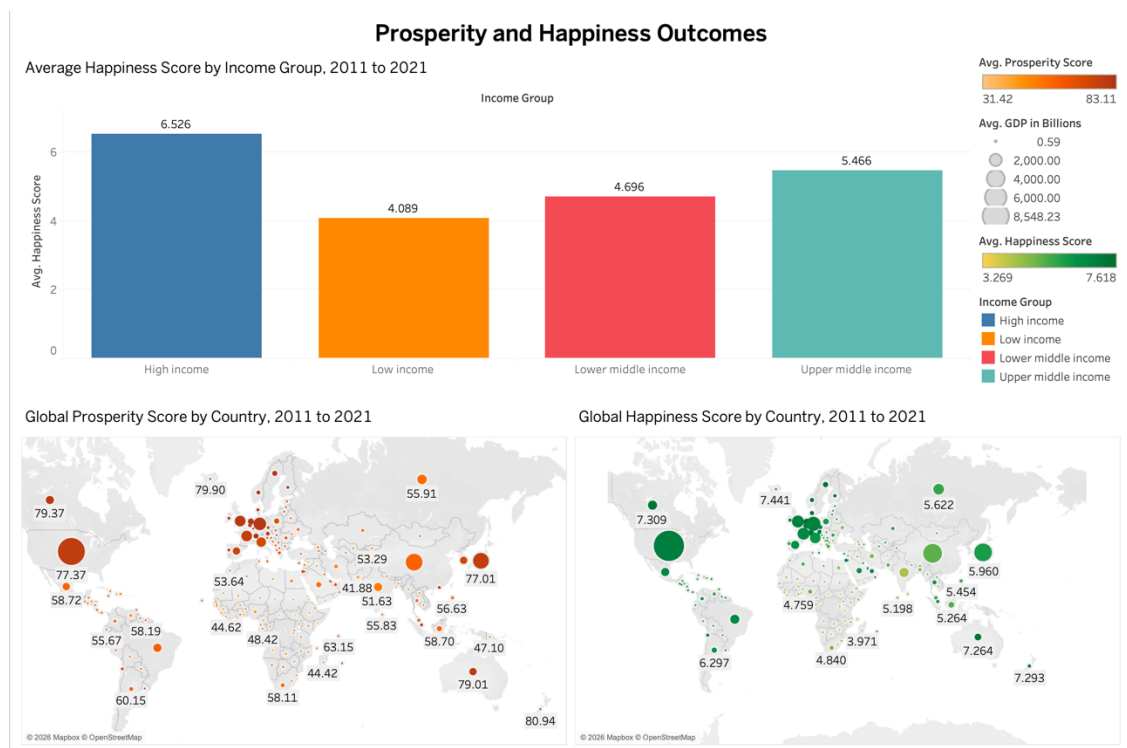


This chart shows the average Happiness Score by World Bank income group from 2011 to 2021. The bar chart compares high income, upper middle income, lower middle income, and low income countries. It helps show how happiness outcomes differ across countries with different income levels.

High income countries have the highest average Happiness Score at 6.526, followed by upper middle income countries at 5.466. Lower middle income countries have an average score of 4.696, while low income countries have the lowest average score at 4.089. This shows a clear pattern where higher income groups generally have higher happiness scores.

This chart supports **Research Question 5** by showing that happiness outcomes vary strongly across income groups. It also shows that income level is closely associated with happiness, but it should not be treated as the only explanation. Other factors such as health, social support, governance, safety, and public services may also influence national happiness outcomes.

Dashboard 5 : Prosperity and Happiness Outcomes



This dashboard explores prosperity and happiness outcomes across countries and World Bank income groups from 2011 to 2021. The three visualizations collectively address Research Question 5:

Which countries stand out as high education spenders, high military spenders, high growth countries, or prosperity and happiness outliers?

It enables users to:

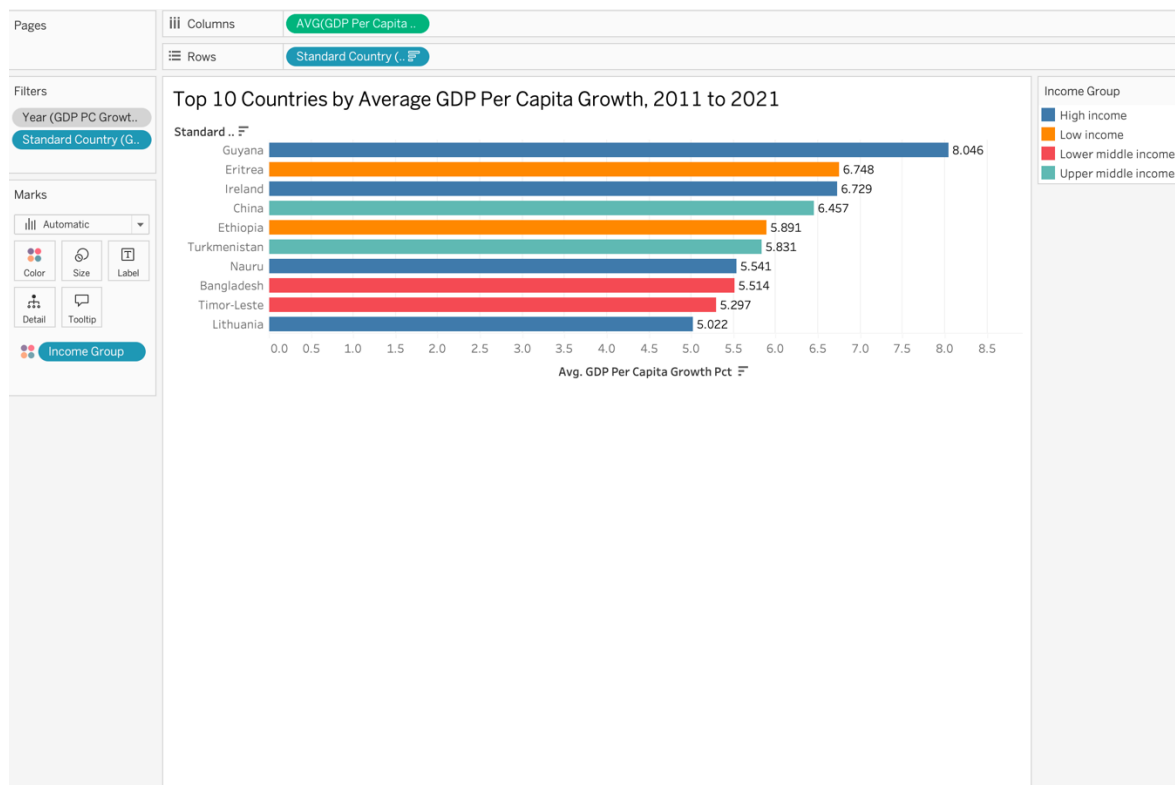
- Compare average Happiness Score across World Bank income groups.
- Identify country-level prosperity patterns using the global prosperity map.
- Identify country-level happiness patterns using the global happiness map.
- Compare whether larger economies also show stronger prosperity and happiness outcomes.

The dashboard shows that high income countries have the highest average Happiness Score, followed by upper middle income countries. Lower middle income and low income countries show lower average happiness scores. This suggests that income level is strongly associated with happiness outcomes, although income alone does not fully explain national well-being.

The prosperity and happiness maps show that countries in North America, Western Europe, Australia, New Zealand, and parts of East Asia generally have stronger prosperity and happiness outcomes. Many countries in parts of Africa, South Asia, and the Middle East show lower scores. The maps also show that large GDP size does not always mean the highest prosperity or happiness score.

Overall, this dashboard supports **Research Question 5** by identifying country-level prosperity and happiness patterns across the world. It shows that some countries perform better or worse than others, even within similar income groups. The dashboard also shows that GDP and spending levels alone cannot fully explain prosperity and happiness. Other factors such as governance, health, social support, safety, and public service quality may also influence these outcomes.

19. Top 10 Countries by Average GDP Per Capita Growth, 2011 to 2021

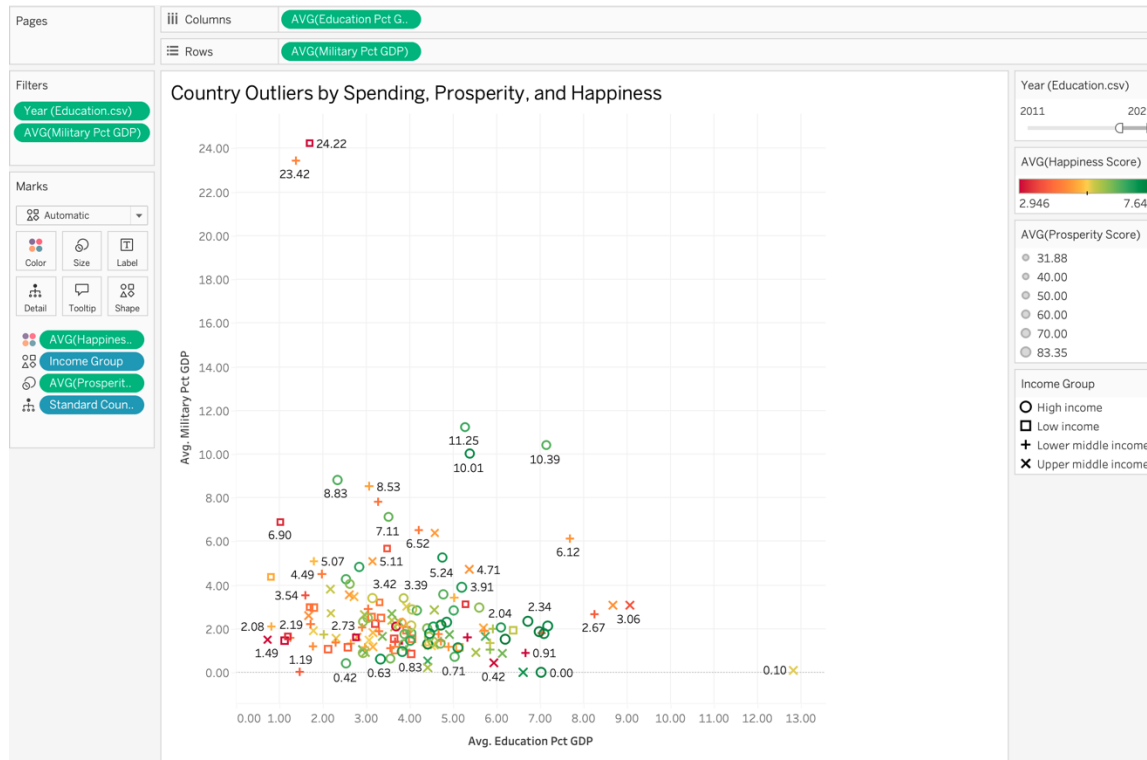


This chart shows the Top 10 countries by average GDP per capita growth from 2011 to 2021. The bar chart compares countries across World Bank income groups and highlights which countries had the strongest average economic growth per person during this period.

Guyana has the highest average GDP per capita growth at 8.046%, followed by Eritrea, Ireland, China, Ethiopia, Turkmenistan, Nauru, Bangladesh, Timor-Leste, and Lithuania. The chart includes countries from different income groups, which shows that high growth is not limited to only high income countries. Some low income, lower middle income, and upper middle income countries also appear among the top growth performers.

This chart supports **Research Question 5** by identifying countries that stand out as high-growth outliers. It shows that some countries achieved strong GDP per capita growth even outside the high income group. However, this chart only shows growth performance, not the causes of growth. Other factors such as natural resources, trade, investment, policy changes, and recovery from earlier economic shocks may also explain these high-growth patterns.

20. Country Outliers by Spending, Prosperity, and Happiness

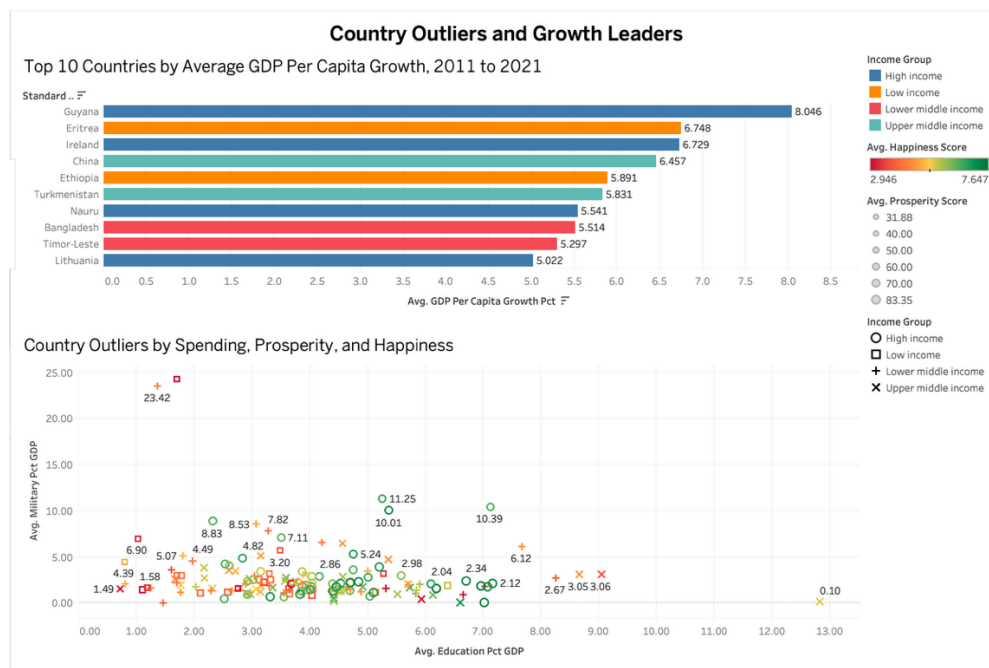


This chart shows country outliers based on education spending, military spending, prosperity, and happiness from 2011 to 2021. The x-axis shows average education spending as a percentage of GDP, and the y-axis shows average military spending as a percentage of GDP. The color shows average Happiness Score, the size shows average Prosperity Score, and the shape shows the World Bank income group.

Most countries are clustered in the lower-left part of the chart, meaning they spend a moderate share of GDP on both education and military expenditure. A few countries stand out clearly as outliers, especially countries with very high military spending above 20% of GDP and countries with very high education spending near the right side of the chart. These outliers show that some countries follow very different spending patterns compared with the majority.

The chart supports **Research Question 5** by identifying countries that stand out in spending, prosperity, and happiness outcomes. It shows that countries with similar income groups can still have different spending priorities and different development outcomes. The chart also shows that spending levels alone do not fully explain prosperity or happiness, because countries with similar spending patterns can still have different outcome scores.

Dashboard 6 : Country Outliers and Growth Leaders



This dashboard explores country-level outliers and growth leaders from 2011 to 2021. The two visualizations collectively address Research Question 5:

Which countries stand out as high education spenders, high military spenders, high growth countries, or prosperity and happiness outliers?

It enables users to:

- Identify the top 10 countries with the highest average GDP per capita growth.
- Compare high-growth countries across World Bank income groups.
- Examine country outliers based on education spending, military spending, prosperity, and happiness.
- Understand whether countries with similar income levels show different spending and development outcome patterns.

The Top 10 growth chart shows that Guyana has the highest average GDP per capita growth, followed by Eritrea, Ireland, China, Ethiopia, Turkmenistan, Nauru, Bangladesh, Timor-Leste, and Lithuania. These countries belong to different income groups, which shows that high GDP per capita growth is not limited to high income countries.

The country outlier chart shows that most countries are clustered in the lower-left area, meaning they spend a moderate share of GDP on both education and military expenditure. A few countries stand out with very high military spending or very high education spending. The chart also uses happiness score, prosperity score, and income group to show that countries with similar spending levels can still have different development outcomes.

Overall, this dashboard supports **Research Question 5** by identifying countries that stand out in growth, spending, prosperity, and happiness. It shows that some countries perform differently from others within the same income group. The dashboard also shows that GDP growth, prosperity, and happiness cannot be explained by spending levels alone. Other factors such as governance, conflict exposure, natural resources, public policy, and economic structure may also influence these outcomes.

CONCLUSION

This research analyzed how education spending, military spending, GDP per capita growth, prosperity, and happiness differ across countries by World Bank income group. Instead of using developed, developing, and underdeveloped categories, the project used World Bank income groups: high income, upper middle income, lower middle income, and low income. This made the country comparisons more consistent and easier to explain.

The results show that high income countries have the highest average GDP, followed by upper middle income countries. Lower middle income and low income countries remain much lower in overall GDP. This shows a clear economic gap across income groups. The GDP per capita growth chart also shows that growth was not stable every year. Most income groups grew from 2011 to 2019, dropped sharply in 2020, and recovered in 2021. This shows that global shocks can affect all income groups, even if the size of the impact differs.

Education spending showed an interesting pattern. Higher income does not always mean a higher percentage of GDP is spent on education. Some lower middle income and upper middle income countries spent a larger share of GDP on education than high income countries. The Top 10 education spending chart also showed that countries such as Micronesia, Marshall Islands, Cuba, Solomon Islands, and Kiribati had very high education spending as a percentage of GDP. This shows that education priority is not limited to wealthy countries.

The relationship between education spending and GDP per capita growth was mixed. Some income groups showed a positive relationship, while others showed weak or uneven patterns. This means higher education spending does not automatically create higher short-term economic growth. However, education spending showed a clearer connection with broader development outcomes such as prosperity and happiness. Countries with stronger education investment often had better prosperity and happiness patterns, but the relationship was still different across income groups.

Military spending also differed across income groups. High income countries showed the highest average military spending as a percentage of GDP for most years. The Top 10 military spending chart showed countries such as Saudi Arabia, Oman, South Sudan, Qatar, and the United Arab Emirates spending the highest share of GDP on military. The distribution chart also showed that military spending varies widely within the same income group. So income level alone does not fully explain military spending behavior.

The relationship between military spending and GDP per capita growth was not consistent. Some income groups showed a positive trend, while others showed a negative trend. This means the project cannot claim that military spending directly increases or decreases economic growth. The results suggest that the connection between military spending and growth depends on other factors such as conflict, governance, security needs, economic stability, and public investment choices.

The education and military trade-off dashboard showed that most countries spend a moderate share of GDP on both education and military expenditure. On average, all income groups spent more on education than military, but the size of the gap differed. Upper middle income countries had the largest education- military gap, while lower middle income countries had the smallest gap. This shows that countries with similar income levels can still make different spending choices.

The prosperity and happiness maps showed clear country-level differences. Countries in North America, Western Europe, Australia, New Zealand, and parts of East Asia generally had stronger prosperity and happiness scores. Many countries in parts of Africa, South Asia, and the Middle East had lower scores. The maps also showed that large GDP size does not always mean the highest prosperity or happiness score.

The project shows that education and military spending are associated with development outcomes, but they do not explain everything by themselves. Education spending appears more closely connected with prosperity and happiness than with short-term GDP per capita growth. Military spending shows mixed patterns and should be interpreted carefully. Income group, governance, public policy, conflict exposure, health systems, infrastructure, and social support also influence national outcomes.

The main conclusion is that balanced spending matters. Countries need military spending for security, but long-term prosperity also depends on investment in education, health, infrastructure, and public services. Increasing education spending alone is not enough. Good governance and effective policy implementation are also needed to turn spending into real development outcomes. The findings should be understood as relationships and patterns, not direct proof of cause and effect.

FUTURE RESEARCH QUESTIONS

- How does the quality of governance affect the relationship between education spending and national prosperity across income groups?
- How does conflict or political instability change the relationship between military spending and GDP per capita growth?
- How do health spending, education spending, and military spending together explain differences in happiness and prosperity outcomes across countries?